

Comparison Theorem

Thm 6.16 in Rotman

Setting:

- Within any Abelian Category

- morphism $f: A \rightarrow A'$

- $\mathcal{P}_A$ is a projective complex, while $\mathcal{P}'_A$ is an exact complex

$$\begin{array}{ccccccc} \cdots & \rightarrow & \mathcal{P}_2 & \xrightarrow{d_2} & \mathcal{P}_1 & \xrightarrow{d_1} & \mathcal{P}_0 \xrightarrow{\varepsilon} A \rightarrow \emptyset & =: \mathcal{P}_A \\ & & \downarrow \check{f}_2 & & \downarrow \check{f}_1 & & \downarrow \check{f}_0 & \downarrow f \\ \cdots & \rightarrow & \mathcal{P}'_2 & \xrightarrow{d'_2} & \mathcal{P}'_1 & \xrightarrow{d'_1} & \mathcal{P}'_0 \xrightarrow{\varepsilon'} A' \rightarrow \emptyset & =: \mathcal{P}'_A \end{array}$$

Statement:

There is a chain map $\check{f}: \mathcal{P}_A \rightarrow \mathcal{P}'_A$ making the diagram at top right commute that is unique up to chain homotopy.

Proof:

We will construct $\check{f}$ via induction, then construct a chain homotopy between $\check{f}$ and any other chain map constructed with f .

① :

Base of Induction $n = \emptyset$:

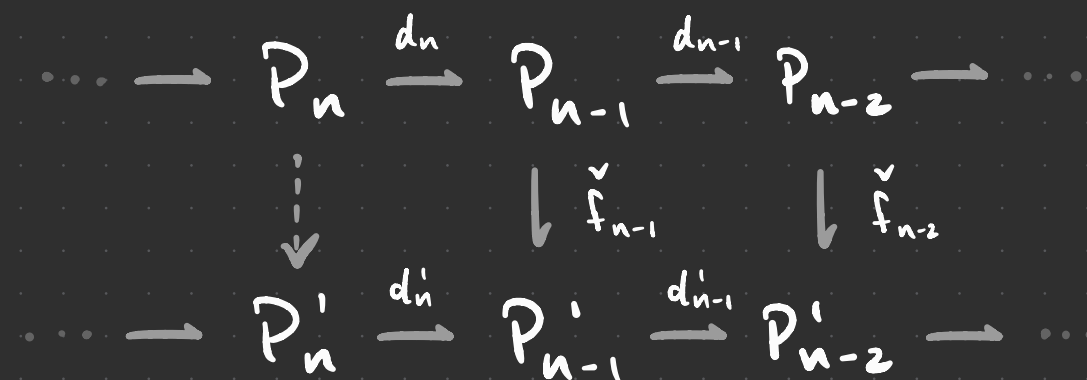
Consider the diagram at right. By the surjectivity of ε' and the projectivity of P_0 , there is a map, call it $\check{f}_0$, from P_0 to P'_0 making the diagram at right commute, which means $\varepsilon' \check{f}_0 = f \varepsilon$.



This completes the base step of induction.

Induction Step:

We first prove that $\text{im}(\check{f}_{n-1}, d_n) \subseteq \text{im } d'_n$, because it will give us a bridge to our desired result, namely the existence of an $\check{f}_n$ that makes the square commute. Note that we define $\check{f}_{-1} = f$ to ensure valid induction.



Note the following:

- by exactness, $\text{im } d'_n = \ker d'_{n-1}$.
- by induction hypotheses, $d'_{n-1} \check{f}_{n-1} = \check{f}_{n-2} d_{n-1}$.

With these facts in mind, we act d'_{n-1} on $\check{f}_{n-1} d_n$ to yield:

$$\begin{aligned} d'_{n-1}(\check{f}_{n-1} d_n) &= d'_{n-1} \check{f}_{n-1} d_n \\ &= \check{f}_{n-2} d_{n-1} d_n \\ &= \check{f}_{n-2}(d_{n-1} d_n) = \emptyset \end{aligned}$$

In the last equality, we used the complex condition on $\mathcal{P}_A$. Thus we've demonstrated that $\text{im}(\check{f}_{n-1} d_n) \subseteq \ker d'_{n-1} = \text{im } d'_n$.

With this image statement we can now write the diagram at left below.

$$\begin{array}{ccc} & \check{f}_n & \\ & \swarrow & \\ P'_n & \xrightarrow{d'_n} & \text{im } d'_n \end{array}$$

Clearly d'_n surjects onto its image, hence the projectivity of P_n gives us the existence of $\check{f}_n$ with $d'_n \check{f}_n = \check{f}_{n-1} d_n$, as desired.

②:

Base of Induction:

Suppose we have another chain map $h: \mathcal{P}_A \rightarrow \mathcal{P}_{A'}$ such that $\varepsilon' h_0 = f \varepsilon$. We define

$s_{-1}: A \rightarrow \mathcal{P}'_0$ and $s_{-2}: \Phi \rightarrow A'$ to be zero maps, and relabel $\varepsilon =: d_0$ and $\varepsilon' =: d'_0$. Further, note that $h_{-1} = f =: \check{f}_{-1}$.

$$\begin{array}{ccccc} \mathcal{P}_0 & \xrightarrow{d_0} & A & \xrightarrow{d_{-1}} & \Phi \\ \check{f}_0 \downarrow & \nearrow s_{-1} & \downarrow \check{f}_{-1} & \nearrow s_{-2} & \\ \mathcal{P}'_0 & \xrightarrow{d'_0} & A' & & \end{array}$$

The chain homotopy condition to check then becomes:

$$\check{f}_{-1} - h_{-1} = d'_0 s_{-1} + s_{-2} d_{-1}$$

Since $\check{f}_{-1}$ and h_{-1} are both just f , the LHS is zero. The definition of s_{-1} and s_{-2} then makes the RHS zero as well, thus verifying the base case of induction.

Induction Step:

We proceed similarly to the induction step for the construction of $\check{f}$ in that we prove a bridging condition, then prove the bridge to the desired result.

$$\begin{array}{ccccccc} & & \mathcal{P}_n & \xrightarrow{d_n} & \mathcal{P}_{n-1} & \xrightarrow{d_{n-1}} & \mathcal{P}_{n-2} \\ & \nwarrow s_n & \downarrow h_n & \downarrow \check{f}_n & \nearrow s_{n-1} & \downarrow & \nearrow s_{n-2} \\ \mathcal{P}'_{n+1} & \xrightarrow{d'_{n+1}} & \mathcal{P}'_n & \xrightarrow{d'_n} & \mathcal{P}'_{n-1} & & \end{array}$$

Our bridging condition to prove in this case is:

$$\text{im}(h_n - \check{f}_n - s_{n-1} d_n) \subseteq \text{im } d_{n+1}'$$

By exactness, $\text{im } d_{n+1}' = \ker d_n'$, so then we consider:

$$d_n' (h_n - \check{f}_n - s_{n-1} d_n) \stackrel{1}{=} d_n' h_n - d_n' \check{f}_n - d_n' s_{n-1} d_n$$

$$\stackrel{2}{=} d_n' h_n - d_n' \check{f}_n - (h_{n-1} - \check{f}_{n-1} - s_{n-2} d_{n-1}) d_n$$

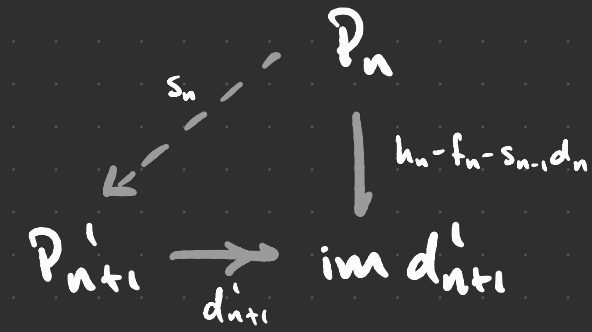
$$\stackrel{3}{=} d_n' h_n - d_n' \check{f}_n - h_{n-1} d_n + \check{f}_{n-1} d_n + s_{n-2} d_{n-1} d_n$$

$$\stackrel{4}{=} d_n' h_n - d_n' \check{f}_n - d_n' h_n + d_n' \check{f}_n + s_{n-2} \emptyset$$

$$= \emptyset$$

In equality 1 and 3 we used the fact that function composition distributes over function addition, whereas in equality 2 we used the $(n-1)^{\text{st}}$ homotopy condition from the induction assumption, and in equality 4 the commutativity resulting from the chain maps h and $\check{f}$.

We now know that $\text{im}(h_n - f_n - s_{n-1}d_n) \subseteq \ker d'_n = \text{im } d'_{n+1}$, and so we can validly construct the following diagram at right:



Then by surjectivity of d'_{n+1} onto its image and the projectivity of P_n , we acquire a map $s_n: P_n \rightarrow P'_{n+1}$ with $d'_{n+1}s_n = h_n - f_n - s_{n-1}d_n$. This is the n^{th} chain

homotopy condition slightly rephrased, thus we've proven the desired result.

□